Case Study 3: Retail Billing with Multiple Discount Rule

1. Problem Statement

To develop a Python program that generates a retail bill for multiple products by considering product prices, quantities, category-specific taxes, discount conditions, and maximum discount limits. The program should display an itemized invoice showing the total amount, discount, tax, and final bill.

2. Objectives

- To calculate the total price of multiple products.
- To apply discounts based on purchase amounts.
- To calculate taxes for different product categories.
- To apply a maximum discount limit.
- To generate an itemized bill with the final payable amount.

3. Assumptions

The following values are assumed for this assignment.

Component	Rule
Electronics tax	18%
Clothing tax	5%
Grocery tax	0%
Discount	10% if subtotal is ₹2,000 or more

Maximum discount	₹500
Discount application	Discount is deducted before calculating tax

These are sample assignment rules, not official tax rates for actual products.

4. Algorithm

1. Start the program.
2. Enter the number of products.
3. For each product, enter its name, category, price, and quantity.
4. Calculate the product subtotal using price multiplied by quantity.
5. Add all product subtotals to calculate the total subtotal.
6. If the subtotal is ₹2,000 or more, calculate a 10% discount.
7. Limit the discount to a maximum of ₹500.
8. Calculate the taxable amount after discount.
9. Calculate tax according to each product's category.
10. Calculate the final bill and display the itemized invoice.
11. Stop the program.

5. Flowchart

Start
Input product details, prices, quantities and categories
Calculate subtotal for all products
Check discount eligibility and apply maximum limit
Calculate category-wise tax after discount
Display itemized invoice and final amount
Stop

6. Pseudocode

```
None
START

INPUT number_of_products
SET subtotal = 0
```

```
SET total_tax = 0

FOR each product:
    INPUT name, category, price, quantity
    product_total = price * quantity
    ADD product_total to subtotal
    STORE product details

IF subtotal >= 2000:
    discount = subtotal * 10 / 100
    IF discount > 500:
        discount = 500
ELSE:
    discount = 0

FOR each product:
    allocated_discount = discount * product_total / subtotal
    taxable_amount = product_total - allocated_discount

    IF category is Electronics:
        tax_rate = 18%
    ELSE IF category is Clothing:
        tax_rate = 5%
    ELSE:
        tax_rate = 0%

    tax = taxable_amount * tax_rate
    ADD tax to total_tax

final_bill = subtotal - discount + total_tax

DISPLAY itemized invoice
DISPLAY subtotal, discount, total_tax, final_bill

STOP
```

7. Complete Python source code

Save the file as `retail_billing.py`.

None

```
# Retail Billing with Multiple Discount Rules

products = []

number_of_products = int(input("Enter number of products: "))

if number_of_products <= 0:
    print("Please enter a valid number of products.")

else:
    valid_categories = ["electronics", "clothing", "grocery"]

    for i in range(number_of_products):
        print("\nProduct", i + 1)

        name = input("Enter product name: ").strip()
        category = input(
            "Enter category (electronics/clothing/grocery): "
        ).strip().lower()
        price = float(input("Enter price: Rs. "))
        quantity = int(input("Enter quantity: "))

        if (not name or category not in valid_categories
                or price < 0 or quantity <= 0):
            print("Invalid product details. Please run again.")
            break

        product_total = price * quantity

        products.append({
            "name": name,
            "category": category,
```

```
            "price": price,
            "quantity": quantity,
            "total": product_total
        })

    else:
        subtotal = sum(product["total"] for product in products)

        # Calculate discount
        if subtotal >= 2000:
            discount = min(subtotal * 0.10, 500)
        else:
            discount = 0

        total_tax = 0

        print("\n========== RETAIL INVOICE ==========")
        print("Product       Qty    Amount")
        print("-----------------------------------")

        for product in products:
            if subtotal > 0:
                share = product["total"] / subtotal
                allocated_discount = discount * share
            else:
                allocated_discount = 0

            taxable_amount = product["total"] - allocated_discount

            if product["category"] == "electronics":
                tax_rate = 0.18
            elif product["category"] == "clothing":
                tax_rate = 0.05
            else:
                tax_rate = 0.00
```

```
            tax = taxable_amount * tax_rate
            total_tax += tax

            print(
                product["name"],
                product["quantity"],
                "Rs.", round(product["total"], 2)
            )

        final_bill = subtotal - discount + total_tax

        print("-------------------------------------")
        print("Subtotal       : Rs.", round(subtotal, 2))
        print("Discount       : Rs.", round(discount, 2))
        print("Total Tax      : Rs.", round(total_tax, 2))
        print("Final Bill     : Rs.", round(final_bill, 2))
        print("=====================================")
```

8. Test cases and expected output

Enter the following sample inputs.

None

```
Enter number of products: 3

Product 1
Enter product name: Headphones
Enter category (electronics/clothing/grocery): electronics
Enter price: Rs. 1500
Enter quantity: 1

Product 2
Enter product name: Shirt
```

```
Enter category (electronics/clothing/grocery): clothing
Enter price: Rs. 800
Enter quantity: 1

Product 3
Enter product name: Rice
Enter category (electronics/clothing/grocery): grocery
Enter price: Rs. 200
Enter quantity: 1
```

Expected calculations:

Product	Subtotal	Discount allocated	Tax
Headphones	₹1,500	₹150	₹243
Shirt	₹800	₹80	₹36
Rice	₹200	₹20	₹0
Total	₹2,500	₹250	₹279

Final bill = ₹2,500 − ₹250 + ₹279 = ₹2,529

Expected final output:

```
None

========== RETAIL INVOICE ==========
Product       Qty    Amount
```

```
-------------------------------------
Headphones    1      Rs. 1500.0
Shirt         1      Rs. 800.0
Rice          1      Rs. 200.0
-------------------------------------
Subtotal       : Rs. 2500.0
Discount       : Rs. 250.0
Total Tax      : Rs. 279.0
Final Bill     : Rs. 2529.0
=====================================
```

9. Output screenshot

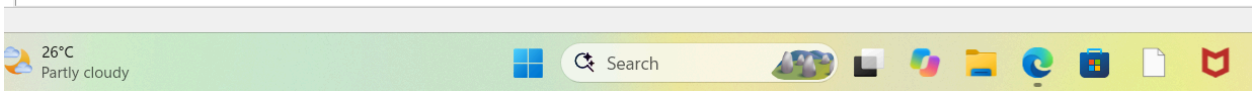
```
======== RESTART: C:/Users/YOGITHA/OneDrive/Desktop/3.Retail billing.py ========
Enter number of products: 3

Product 1
Enter product name: Headphones
Enter category (electronics/clothing/grocery): electronics
Enter price: Rs. 1500
Enter quantity: 1

Product 2
Enter product name: Shirt
Enter category (electronics/clothing/grocery): clothing
Enter price: Rs. 800
Enter quantity: 1

Product 3
Enter product name: Rice
Enter category (electronics/clothing/grocery): grocery
Enter price: Rs. 200
Enter quantity: 1

========== RETAIL INVOICE ==========
Product       Qty    Amount
-------------------------------------
Headphones 1 Rs. 1500.0
Shirt 1 Rs. 800.0
Rice 1 Rs. 200.0
-------------------------------------
Subtotal       : Rs. 2500.0
Discount       : Rs. 250.0
Total Tax      : Rs. 279.0
Final Bill     : Rs. 2529.0
=====================================
>
```



```python
products = []

number_of_products = int(input("Enter number of products: "))

if number_of_products <= 0:
    print("Please enter a valid number of products.")

else:
    valid_categories = ["electronics", "clothing", "grocery"]

    for i in range(number_of_products):
        print("\nProduct", i + 1)

        name = input("Enter product name: ").strip()
        category = input(
            "Enter category (electronics/clothing/grocery): "
        ).strip().lower()
        price = float(input("Enter price: Rs. "))
        quantity = int(input("Enter quantity: "))

        if (not name or category not in valid_categories
                or price < 0 or quantity <= 0):
            print("Invalid product details. Please run again.")
            break

        product_total = price * quantity

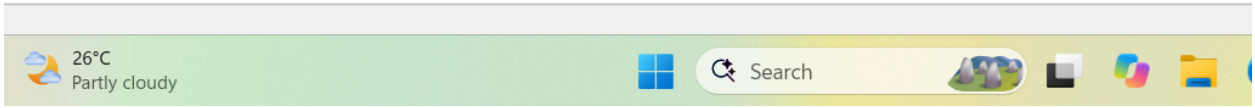
        products.append({
            "name": name,
            "category": category,
            "price": price,
            "quantity": quantity,
            "total": product_total
        })

    else:
        subtotal = sum(product["total"] for product in products)

        # Calculate discount
        if subtotal >= 2000:
            discount = min(subtotal * 0.10, 500)
        else:
            discount = 0

        total_tax = 0

        print("\n========== RETAIL INVOICE ==========")
        print("Product       Qty    Amount")
        print("-------------------------------------")
```



```
        })

    else:
        subtotal = sum(product["total"] for product in products)

        # Calculate discount
        if subtotal >= 2000:
            discount = min(subtotal * 0.10, 500)
        else:
            discount = 0

        total_tax = 0

        print("\n========== RETAIL INVOICE ==========")
        print("Product       Qty    Amount")
        print("-------------------------------------")

        for product in products:
            if subtotal > 0:
                share = product["total"] / subtotal
                allocated_discount = discount * share
            else:
                allocated_discount = 0

            taxable_amount = product["total"] - allocated_discount

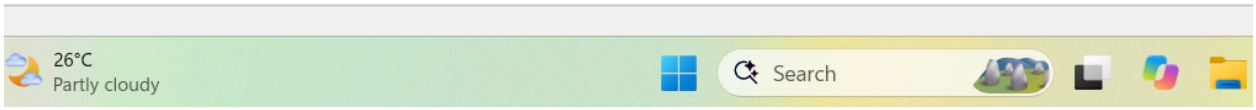
            if product["category"] == "electronics":
                tax_rate = 0.18
            elif product["category"] == "clothing":
                tax_rate = 0.05
            else:
                tax_rate = 0.00

            tax = taxable_amount * tax_rate
            total_tax += tax

            print(
                product["name"],
                product["quantity"],
                "Rs.", round(product["total"], 2)
            )

        final_bill = subtotal - discount + total_tax

        print("-------------------------------------")
        print("Subtotal       : Rs.", round(subtotal, 2))
        print("Discount       : Rs.", round(discount, 2))
        print("Total Tax      : Rs.", round(total_tax, 2))
        print("Final Bill     : Rs.", round(final_bill, 2))
        print("=====================================")
```



10. Conclusion

The Retail Billing System calculates the cost of multiple products, applies discount rules, calculates category-wise taxes, and generates an itemized invoice. It demonstrates the use of Python variables, operators, conditional statements, loops, lists.